

From Representation Credit to Cellular Credit: A Chronological Audit of pART-Inspired Models of Dendritic Vectorization

Repository-based research manuscript
Based on the `OgnjenX/part-vector-credit` experimental record

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Abstract

How can a delayed scalar behavioral outcome support the cell-specific changes required for a brain-computer interface (BCI) in which intermingled cortical neurons have opposite causal roles? This manuscript reports a chronological analysis of the `part-vector-credit` repository. The work was motivated by dendritic signals reported during BCI learning by Francioni et al. [Francioni et al., 2026] and asks whether pART/SMART-inspired mechanisms can account for such signals without an explicit neuron-wise error vector. Repository-derived results are separated throughout from claims about published theory and biology. A prestructured opponent-code routing study (EXP000) showed only that an existing pattern can be selectively expressed. Random-repertoire causal selection, learned top-down expectancy, and an integrated pART-SMART composition (EXP001-EXP003b) established that delayed scalar reinforcement can select useful distributed hypotheses, maintain them across delay, and learn or remap patterned expectations, but did not meet the frozen longitudinal dendritic criterion. A topology-origin study (EXP004) showed that the learned expectation was principally a compression of outcome-selected responses from a fixed repertoire, not construction of arbitrary cellular topology. A source audit and generic local-eligibility diagnostic (EXP005) found no published Grossberg equation that performs the missing operation. The separately labeled generic three-factor comparator nevertheless learned arbitrary hidden sign maps (at $N = 32$, alignment 0.987 after acquisition and 0.964 after remapping). Thus scalar outcome is computationally sufficient in this abstract task when coupled to a local eligibility trace; the repository does not establish that the brain uses that rule, that Grossberg’s published theory contains it, or that Francioni’s dendritic signal is an error gradient.

1 Introduction

Neural learning faces both a temporal and a structural credit-assignment problem: a consequence may arrive after the relevant action, and many simultaneously active cells may have different causal contributions. In a BCI preparation, Francioni et al. reported role-dependent dendritic signals in intermingled cortical neurons and interpreted them as vectorized instructive signals [Francioni et al., 2026]. A literal reading invites a neuron-indexed teaching vector. An alternative is that common scalar outcome can interact with local state, history, or eligibility to produce cell-specific updates.

The repository studied here tested this alternative at progressively stronger levels. It began with a routing sanity check, removed prewired identity shortcuts, introduced learned top-down expectations, validated a local SMART-derived plasticity motif, audited the source theory, tested whether learned expectations synthesize new topology, and finally ran a generic local-eligibility diagnostic. The central question is deliberately narrow: *which claims are supported by the implemented experiments, and which remain hypotheses about biology?*

2 Related work and theoretical commitments

ART addresses stability–plasticity through category matching, vigilance, reset/search, and learned top-down expectations [Carpenter and Grossberg, 1987]. pART emphasizes structural and temporal credit: a representation can be selected, maintained over a delay, and reinforced by broad motivational signals [Grossberg, 2018]. These mechanisms naturally motivate representation-level credit, but do not by themselves specify an arbitrary neuron-resolved BCI feedback vector.

SMART supplies a more cellular circuit motif. In a visual thalamocortical setting, learned higher-to-lower feedback reaches apical dendrites, a match-dependent on-center/off-surround circuit changes timing and participation, and a local voltage/conductance-gated rule changes synapses [Grossberg and Versace, 2008]. Canonical laminar extensions describe related modulatory feedback motifs [Grossberg, 2021]. These sources support a chain from matched feedback to local plasticity, not a demonstrated mapping from delayed BCI reward to arbitrary intermingled P+/P- roles. Grossberg’s vector associative motor systems use an available target-versus-present difference vector [Gaudiano and Grossberg, 1991]; importing that vector would change the information boundary of the present task.

Three alternatives should therefore remain distinct. First, a centralized system might compute and route an explicit cell-wise error. Second, a representation-level system can select one useful pre-existing distributed pattern. Third, a local exploratory rule could use the same scalar advantage at every cell but multiply it by a cell-specific eligibility. Only the third can construct arbitrary signs in the present abstraction without a supplied vector, and it is not attributed to Grossberg in this manuscript.

3 Methods: common task, information boundary, and adjudication

The experiments use abstract closed-loop BCI tasks with hidden, balanced causal roles $c_i \in \{-1, +1\}$ assigned to intermingled cells. The learner receives observable context/state, its own activity and local variables, and scalar outcome; c is environment-only except in explicitly labeled oracle controls and offline analysis. Primary endpoints include BCI performance, correlation/alignment of learned patterns with c , remapping after an independently changed hidden map, and interventions that remove proposed causal ingredients.

The theory-led models distinguish a selected category–hypothesis representation $H_{k,h}$, a learned expectation $T_{k,h,i}$, a modeled top-down/apical quantity D_i , plastic lower weights W_i , and somatic output S_i . Their intended longitudinal test is

$$D_{\text{residual},i}^{\text{early}} \longrightarrow \Delta W_i \longrightarrow \Delta S_i, \quad (1)$$

evaluated within a held hypothesis so that simple switching cannot create a spurious effect. SMART-derived models implement the qualitative local dependency

$$\Delta W_{jk} \propto f_G(V_k, \bar{g}_{jk}) [\bar{g}_{jk} f_N(V_k)(\hat{w} - \check{w}) + w_0 - w_{jk}], \quad (2)$$

where conductance, voltage, and spike timing determine local change [Grossberg and Versace, 2008]. This is not a reward-times-neuron-vector rule.

The intentionally non-Grossberg local-eligibility diagnostic (EXP005) instead updates an adaptive cell-wise projection using

$$\Delta A_i = \eta_N (R - b) e_i, \quad (3)$$

Repository ID	Paper-facing study name	Repository classification
EXP000	Prestructured opponent-code routing	Routing sanity check; not de novo cellular credit.
EXP001	Random-repertoire causal selection	Representation selection, not discovery of cell signs.
EXP002	Learned top-down expectancy	Outcome C for the tested composition.
EXP003–3b	SMART motif and integrated pART–SMART composition	Motif validated; integrated frozen composition Outcome C.
EXP004	Origin of neuron-specific topology	Weak-form support only; fixed-repertoire limited.
EXP005	Source audit and local-eligibility diagnostic	Source outcome E; generic diagnostic positive.

Table 1: Chronology and claim boundary. “Outcome C” and “E” are the repository’s classifications, not broad falsifications of theories.

where $R - b$ is a common scalar reward advantage, e_i is a retained local exploratory perturbation eligibility, and η_N corrects the known population-mean scaling. The hidden roles do not enter this update. Exact replay of legal variables was an explicit audit endpoint.

The repository uses frozen held-out seed sets, preregistered effect floors in later experiments, seed-level bootstrap intervals, positive controls supplied with hidden vector information, and ablations such as shuffled outcome, removed exploration, or removed eligibility. These choices make the reported claims conditional on the specified abstractions rather than claims of biological necessity.

4 Results in experimental chronology

4.1 Prestructured opponent-code routing (EXP000)

After two retained failed calibrations, the corrected routing study achieved late accuracy 0.978 across 30 seeds, with P+ modulation +0.673, P- modulation −0.673, and opposition index +1.346. Yet the audit identified a decisive shortcut: hidden P+/P- identity was visible in even/odd indices and already encoded in category templates. A frozen-template reanalysis with category and value learning rates set to zero retained 0.978 late accuracy. Ordinary corrected trials also had 100% resonance and no resets. The routing study therefore supports only an existence claim: selective feedback can route a *pre-existing* opponent pattern without an explicit error vector. It neither learned the signs nor validated reset/search under frequent mismatch.

4.2 Random-repertoire causal selection (EXP001)

The random-repertoire study removed visible population identity and used a random hypothesis bank. The selected distributed hypothesis could be held across irrelevant events, credited by delayed scalar outcome, and relearned after hidden roles changed. This demonstrates structural and temporal credit at the level of selecting a representation. Its limit was equally clear: the useful direction already occurred by chance in the bank; increasing bank capacity increased the probability of a well-aligned direction. Instantaneous dendritic residual alignment also appeared in non-learning controls and did not reliably predict later cell-specific activity change. The explicit neuron-indexed positive control did. Thus the experiment learned *which pattern works*, not which individual cells ought to change sign.

4.3 Learned top-down expectancy (EXP002)

The learned-expectancy study added a Grossberg outstar-derived association, $T_h \leftarrow T_h + \eta(S_{\sim} - T_h)$, to store soma patterns recurrently coactive with selected hypotheses. On frozen data, scalar selection learned behavior (0.528 before and 0.397 after remapping) and role-aligned top-down patterns (0.343 and 0.287). However, direct copying of the chosen pattern produced essentially the same alignment, a contextual bandit matched behavior, and the pre/post-remap longitudinal residual effects were -0.326 and -0.016 . The explicit neuron-indexed control gave 0.660/0.545. Suppressing apical learning erased learned pattern alignment without changing behavior. The tested selection-plus-pattern-completion account therefore failed the requested cellular longitudinal criterion; it does not show that all Grossberg circuits fail.

4.4 SMART motif and integrated pART–SMART composition (EXP003a–b)

The local-mechanism program began with a theory audit (EXP003), not a simulation result. It found that SMART provides a defensible source-derived local arrow: supplied matched feedback can alter spike timing, gate local plasticity, and change future response. The reduced SMART-motif study (EXP003a) reported an update increase from 0.600 to 0.731 and a learning-off feedforward probe changing from 0 to 7 of 8 spikes. Competitor, top-down-ablated, and mismatch conditions had zero update. This validates a narrow local mechanism, not a full SMART reproduction or behavioral credit solution.

The integrated composition (EXP003b) combined the locked motif with delayed structural credit and learned expectations. Across 12 held-out mappings, useful remappable expectations emerged and top-down feedback advanced local spike timing by 6.29 ms, changing lower weights. Nevertheless, early soma-conditioned dendritic residuals did not predict lower-weight or held-hypothesis somatic change at the frozen decision threshold; only the explicitly neuron-wise positive control completed the $D \rightarrow W \rightarrow S$ assay. Post-hoc failure localization did not alter this outcome. It found that raw T and more direct modeled quantities predicted sparse ΔW , whereas the soma-conditioned residual lost much of that information. For example, raw $T \rightarrow \Delta W$ correlation was 0.416 [0.290, 0.530], but residual $D \rightarrow \Delta W$ was -0.004 [-0.141 , 0.128]. Sparse, quantized engagement prevents reclassifying this as confirmatory success.

4.5 Bridge audit

The repository’s bridge audit separates explicit source content from new assumptions. It attributes representation selection/maintenance to pART, expectation learning and match/reset to ART, apical feedback and timing-dependent local plasticity to SMART, and the BCI dendritic statistic to Francioni et al. It identifies the weakest missing arrow as acquisition of arbitrary neuron-fine $H \rightarrow T$ topology from scalar outcome (assumption A2). It also flags unproven mapping of SMART surround to NDNF-sensitive dendritic signals and the observation model from current/voltage to calcium residual. These are productive hypotheses, not settled source-derived consequences.

4.6 Origin of neuron-specific topology (EXP004)

The topology-origin study prospectively tested the origin of T using 16 held-out seeds and 58 frozen scenarios. At fixed bank size $M = 16$, controlled low/medium/high initial coverage produced primary behavior 0.765/0.882/0.962 and T alignment 0.258/0.327/0.395; high-minus-low behavior was 0.197 [0.137, 0.255]. Across random banks, candidate coverage increased with M , but learning

became search-limited. A specially constructed sequential condition did show behavioral composition: primary behavior 0.980 exceeded the best repeated-single action by 0.313 [0.260, 0.374], yet a contextual bandit achieved 0.977.

The representational result was decisive for the implemented rule:

$$\text{corr}(T, \text{selected initial soma}) = 0.999851, \quad \text{corr}(T, \text{mean target}) = 0.999870. \quad (4)$$

Exact replay reconstructed T with RMSE 0.0. T did not exceed the best individual experienced target. Outcome permutation changed topology by altering which hypotheses and targets were visited, not materially by reweighting a fixed visitation history. Hence scalar outcome genuinely selects and consolidates distributed states, but the neuron coordinates arise mainly from fixed repertoire geometry. This is weak-form A2 support, not arbitrary topology construction.

4.7 Source audit and local-eligibility diagnostic (EXP005)

The source audit first imposed a gate: no Grossberg primary run would be invented if no audited source supplied the required operation. It classified the source result as E: pART, ART/outstar, exploration, SMART, and vector motor systems each supply adjacent functions, but no equation joined an independent local exploratory trace to delayed scalar consequence to learn arbitrary intermingled signs.

The separately labeled generic comparator then tested that missing operation. On 16 held-out maps, after 1,280 acquisition episodes and 2,560 remap episodes, final role alignment was 0.997, 0.993, 0.987, and 0.972 for $N = 8, 16, 32, 64$; post-remap alignment was 0.968, 0.960, 0.964, and 0.950. At $N = 32$, shuffled outcome reduced post-remap alignment to 0.229, while no plasticity, no temporal eligibility, and no exploration left alignment near zero; generic-minus-shuffled was 0.735 [0.655, 0.820]. Changed-cell sign reversal was perfect through $N = 32$ and 0.975 at $N = 64$. Legal replay reconstructed updates to numerical precision. During acquisition, final topology exceeded the best single exploratory sample at every N , supporting accumulation rather than lucky fixed-pattern selection.

These results support a computational statement: common scalar advantage plus private local exploratory eligibility can identify arbitrary cellular topology in this abstract task. They do not convert the generic rule into a published Grossberg mechanism, identify a neuronal implementation, or establish the interpretation of dendritic calcium signals.

5 Discussion

The experiments sharpen a useful distinction. Representation-level credit is comparatively easy: a scalar consequence can reinforce a maintained hypothesis and select useful patterns from a repertoire. Cellular credit is harder because it requires a cell-varying quantity. Across the studies of random repertoire selection, learned expectancy, integrated composition, and topology origin (EXP001–EXP004), that quantity was largely supplied by existing population geometry and stored by selection/outstar-like averaging. In the local-eligibility diagnostic (EXP005), it arose from each cell’s own perturbation eligibility; the scalar reward only determined whether that private deviation was reinforced or opposed.

The result does not require a central vector-error broadcaster in principle. But “not mathematically necessary” is not evidence that biology implements node perturbation. A dendritic signal that looks neuron-specific could reflect explicit routed instruction, locally computed credit, selected expectation, match gating, or a mixture. The failed frozen longitudinal residual test and its post-hoc

localization further show that an observation model can obscure information present in currents or timing, especially in sparse regimes. That is a measurement issue to test prospectively, not a license to rescue a failed confirmatory endpoint.

6 Limitations

The models are abstract BCI simulations, not reconstructions of RSC circuitry, animal behavior, or full pART/SMART equations. Several key bridges are engineering extrapolations: binding a pART-selected representation to a SMART-like projection, targeting the relevant BCI cells, mapping center/surround to NDNF circuitry, and mapping modeled activity to a soma-conditioned calcium residual. The prestructured routing study (EXP000) is structurally favored and should never be used as de novo credit evidence. The integrated composition (EXP003b) used sparse, quantized local plasticity; post-hoc analyses are diagnostic only. The topology-origin study (EXP004) establishes fixed-repertoire dependence only for its specified bank and outstar ownership rule. The local-eligibility diagnostic’s scaling correction (EXP005) is known-task normalization, not role information, but finite-task success remains distinct from biological scalability. Finally, repository provenance includes companion worktree/branch records for EXP004–EXP005; readers should verify hashes and source history before treating the synthesis as a formal release.

7 Biological predictions and discriminating tests

The combined record suggests testable, mutually distinguishing predictions.

1. **Eligibility–outcome interaction.** A local-credit account predicts that spontaneous or imposed cell-specific pre-outcome deviations leave a trace whose interaction with later scalar outcome predicts signed plasticity. Decorrelating trace and reinforcement timing should reduce learning without necessarily removing scalar reward responses.
2. **Within-cell remapping.** If category/source feedback owns the pattern, changing context or hidden BCI contingencies should reorganize feedback before later somatic change and permit the same cell to reverse sign across contexts.
3. **Center/surround measurement.** Simultaneous measurement and independent perturbation of excitatory feedback and NDNF-associated inhibitory surround should test whether a signed local membrane influence, rather than an excitatory calcium proxy alone, carries the predictive signal.
4. **Projection- and phase-specific intervention.** Perturb a candidate feedback source selectively before outcome while preserving sensory error and reward. Loss of learning-dependent role information would support a causal source; a broad activity suppression would be less diagnostic.

These experiments can separate a routed-vector account, a representation-selection account, and an exploratory eligibility account more cleanly than further qualitative matching to a dendritic residual.

8 Conclusion

The repository supports a disciplined two-level conclusion. pART/ART/SMART-inspired mechanisms can plausibly provide delayed representation credit, learned expectations, local match-dependent plasticity, and structured feedback. In the tested compositions they did not establish the missing arbitrary neuron-wise credit operation or the frozen Francioni-like longitudinal residual

signature. A generic three-factor rule did solve that operation using only a global scalar advantage and cell-local eligibility. This establishes computational sufficiency under explicit new assumptions, not a biological mechanism or a source-derived Grossberg explanation. The next decisive work is to measure the proposed local eligibility and feedback bridges directly, with preregistered observation models and causal perturbations.

Data and code availability

All computational results reported here derive from the archived protocols, frozen analyses, and held-out experiment records accompanying this study. Post-hoc analyses are explicitly identified and do not alter the original frozen classifications. Claims derived from the computational experiments are distinguished from interpretations based on outside literature. No new simulation results were generated specifically for this manuscript. The public project repository provides the corresponding source, experiment records, and analysis materials.

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